

Center Based Clustering: A Foundational Perspective

Pranjal Awasthi and Maria-Florina Balcan
Princeton University and Carnegie Mellon University

November 10, 2014

Abstract

In the first part of this chapter we detail center based clustering methods, namely methods based on finding a “best” set of center points and then assigning data points to their nearest center. In particular, we focus on k -means and k -median clustering which are two of the most widely used clustering objectives. We describe popular heuristics for these methods and theoretical guarantees associated with them. We also describe how to design worst case approximately optimal algorithms for these problems. In the second part of the chapter we describe recent work on how to improve on these worst case algorithms even further by using insights from the nature of real world clustering problems and data sets. Finally, we also summarize theoretical work on clustering data generated from mixture models such as a mixture of Gaussians.

1 Approximation algorithms for k -means and k -median

One of the most popular approaches to clustering is to define an objective function over the data points and find a partitioning which achieves the optimal solution, or an approximately optimal solution to the given objective function. Common objective functions include center based objective functions such as k -median and k -means where one selects k center points and the clustering is obtained by assigning each data point to its closest center point. Here closeness is measured in terms of a pairwise distance function $d()$, which the clustering algorithm has access to, encoding how dissimilar two data points are. For instance, the data could be points in Euclidean space with $d()$ measuring Euclidean distance, or it could be strings with $d()$ representing an edit distance, or some other dissimilarity score. For mathematical convenience it is also assumed that the distance function $d()$ is a metric. In k -median clustering the objective is to find center points $c_1, c_2, \dots, c_k$, and a partitioning of the data so as to minimize $\Phi_{k\text{-median}} = \sum_x \min_i d(x, c_i)$. This objective is historically very useful and well studied for facility location problems [16, 43]. Similarly the objective in k -means is to minimize $\Phi_{k\text{-means}} = \sum_x \min_i d(x, c_i)^2$. Optimizing this objective is closely related to fitting the maximum likelihood mixture model for a given dataset. For a given set of centers, the optimal clustering for that set is obtained by assigning each data point to its closest center point. This is known as the Voronoi partitioning of the data. Unfortunately, exactly optimizing the k -median and the k -means objectives is a notoriously hard problem. Intuitively this is expected since the objective function is a non-convex function of the variables involved. This apparent hardness can also be formally justified by appealing to the

notion of NP completeness [43, 33, 8]. At a high level the notion of NP completeness identifies a wide class of problems which are in principle equivalent to each other. In other words, an efficient algorithm for exactly optimizing one of the problems in the class on all instances would also lead to algorithms for all the problems in the class. This class contains many optimization problems that are believed to be hard¹ to exactly optimize in the worst case and not surprisingly, k -median and k -means also fall into the class. Hence it is unlikely that one would be able to optimize these objectives exactly using efficient algorithms. Naturally, this leads to the question of recovering approximate solutions and a lot of the work in the theoretical community has focused on this direction [16, 11, 29, 34, 43, 47, 48, 57, 20]. Such works typically fall into two categories, a) providing formal worst case guarantees on all instances of the problem, and b) providing better guarantees suited for nicer, stable instances. In this chapter we discuss several stepping stone results in these directions, focusing our attention on the k -means objective. A lot of the the ideas and techniques mentioned apply in a straightforward manner to the k -median objective as well. We will point out crucial differences between the two objectives as and when they appear. We will additionally discuss several practical implications of these results.

We will begin by describing a very popular heuristic for the k -means problem known as Lloyd’s method. Lloyd’s method [51] is an iterative procedure which starts out with a set of k seed centers and at each step computes a new set of centers with a lower k -means cost. This is achieved by computing the Voronoi partitioning of the current set of centers and replacing each center with the center of the corresponding partition. We will describe the theoretical properties and limitations of Lloyd’s method which will also motivate the need for good worst case approximation algorithms for k -means and k -median. We will see that the method is very sensitive to the choice of the seed centers. Next we will describe a general method based on local search which achieves constant factor approximations for both the k -means and the k -median objectives. Similar to Lloyd’s method, the local search heuristic starts out with a set of k seed centers and at each step swaps one of the centers for a new one resulting in a decrease in the k -means cost. Using a clever analysis it can be shown that this procedure outputs a good approximation to the optimal solution [47]. This is interesting, since as mentioned above, optimizing the k -means is NP-complete, in fact it is NP-complete even for $k = 2$, for points in the Euclidean space [33]².

In the second part of the chapter we will describe some of the recent developments in the study of clustering objectives. These works take a non-worst case analysis approach to the problem. The basic theme is to design algorithms which give good solutions to clustering problems only when the underlying optimal solution has a meaningful structure. We will call such clustering instances as *stable* instances. We would describe in detail two recently studied notions of stability. The first one called *separability* was proposed by Ostrovsky et. al [57]. According to this notion a k -clustering instance is stable if it is much more expensive to cluster the data using $(k - 1)$ or fewer clusters. For such instances Ostrovsky et. al show that one can design a simple Lloyd’s type algorithm which achieves a constant factor approximation. A different notion called *approximation stability* was proposed by Balcan et. al [20]. The motivation comes from the fact that often in practice optimizing an objective function acts as a proxy for the real problem of getting close to the correct unknown ground truth clustering. Hence it is only natural to assume that any good approximation to the proxy function such as k -means or k -median will also be close to the ground truth clustering in

¹This is the famous P vs NP problem, and there is a whole area called Computational Complexity Theory that studies this and related problems [12].

²If one restricts centers to be data points, then k -means can be solved optimally in time $O(n^{k+1})$ by trying all possible k -tuples of centers and choosing the best. The difficulty of k -means for $k = 2$ in Euclidean space comes from the fact that the optimal centers need not be data points.

terms of structure. Balcan et. al show that under this assumption one can design algorithms that solve the end goal of getting close to the the ground truth clustering. More surprisingly this is true even in cases where it is NP -hard to achieve a good approximation to the proxy objective.

In the last part of the chapter we briefly review existing theoretical work on clustering data generated from mixture models. We mainly focus on Gaussian Mixture Models (GMM) which are the most widely studied distributional model for clustering. We will study algorithms for clustering data from a GMM under the assumption that the mean vectors of the component Gaussians are well separated. We will also see the effectiveness of spectral techniques for GMMs. Finally, we will look at recent work on estimating the parameters of a Gaussian mixture model under minimal assumptions.

2 Lloyd's method for k -means

Consider a set A of n points in the d -dimensional Euclidean space. We start by formally defining Voronoi partitions.

Definition 1 (Voronoi Partition). *Given a clustering instance $\mathcal{C} \subset \mathbb{R}^d$ and k points $c_1, c_2, \dots, c_k$, a Voronoi partitioning using these centers consists of k disjoint clusters. Cluster i consists of all the points $x \in \mathcal{C}$ satisfying $d(x, c_i) \leq d(x, c_j)$ for all $j \neq i$.³*

Lloyd's method, also known as the k -means algorithm is the most popular heuristic for k -means clustering in the Euclidean space which has been shown to be one of the top ten algorithms in data mining [69]. The method is an iterative procedure which is described below.

Algorithm LLOYD'S METHOD

1. **Seeding:** Choose k seed points $c_1, c_2, \dots, c_k$. Set $\Phi_{old} = \infty$. Compute the current k -means cost Φ using seed points as centers, i.e.

$$\Phi_{curr} = \sum_{i=1}^n \min_j d^2(x_i, c_j)$$

2. **While** $\Phi_{curr} < \Phi_{old}$,
 - (a) **Voronoi partitioning:** Compute the Voronoi partitioning of the data based on the centers $c_1, c_2, \dots, c_k$. In other words, create k clusters $C_1, C_2, \dots, C_k$ such that $C_i = \{x : d(x, c_i) \leq \min_{j \neq i} d(x, c_j)\}$. Break ties arbitrarily.
 - (b) **Reseeding:** Compute new centers $\hat{c}_1, \hat{c}_2, \dots, \hat{c}_k$, where $\hat{c}_j = \text{mean}(C_j) = \frac{1}{|C_j|} \sum_{x \in C_j} x$. Set $\Phi_{old} = \Phi_{curr}$. Update the current k -means cost Φ_{curr} using the new centers.
 - (c) **Output:** The current set of centers $c_1, c_2, \dots, c_k$.

³Ties can be broken arbitrarily.

We would like to stress that although Lloyd’s method is popularly known as the k -means algorithm, there is a difference between the underlying k -means objective (which is usually hard to optimize) and the k -means algorithm which is a heuristic to solve the problem. An attractive feature of Lloyd’s method is that the k -means cost of the clustering obtained never increases. This follows from the fact that for any set of points, the 1-means cost is minimized by choosing the mean of the set as the center. Hence for any cluster C_i in the partitioning, choosing $\text{mean}(C_i)$ will never lead to a solution of higher cost. Hence if we repeat this method until there is no change in the k -means cost, we will reach a local optimum of the k -means cost function in finite time. In particular the number of iterations will be at most $n^{O(kd)}$ which is the maximum number of Voronoi partitions of a set of n points in $\mathbb{R}^d$ [42]. The basic method mentioned above leads to a class of algorithms depending upon the choice of the seeding method. A simple way is to start with k randomly chosen data points. This choice however can lead to arbitrarily bad solution quality as shown in Figure 1. In addition it is also known that the Lloyd’s method can take upto 2^n iterations to converge even in 2 dimensions [14, 66].

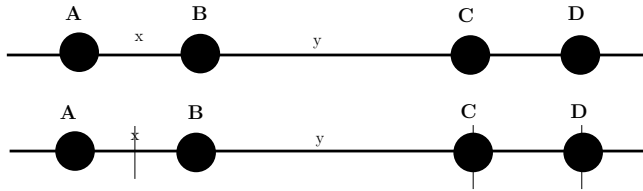


Figure 1: Consider 4 points $\{A, B, C, D\}$ on a line separated by distances x, y and z such that $z < x < y$. Let $k = 3$. The optimal solution has centers at A, B and the centroid of C, D with a total cost of $\frac{z^2}{2}$. When choosing random seeds, there is a constant probability that we choose $\{A, C, D\}$. In this case the final centers will be C, D and the centroid of A, B with a total cost of $\frac{x^2}{2}$. This ratio can be made arbitrarily bad.

In sum, from a theoretical standpoint, k -means with random/arbitrary seeds is not a good clustering algorithm in terms of efficiency or quality. Nevertheless, the speed and simplicity of k -means are quite appealing in practical applications. Therefore, recent work has focused on improving the initialization procedure: deciding on a better way to initialize the clustering dramatically changes the performance of the Lloyd’s iteration, both in terms of quality and convergence properties. For example, [15] showed that choosing a good set of seed points is crucial and if done carefully can itself be a good candidate solution without the need for further iterations. Their algorithm called k -means++ uses the following seeding procedure: it selects only the first center uniformly at random from the data and each subsequent center is selected with a probability proportional to its contribution to the overall error given the previous selections. See Algorithm KMEANS++ for a formal description:

Algorithm KMEANS++

1. **Initialize:** a set S by choosing a data point at random.
2. **While** $|S| < k$,
 - (a) Choose a data point x with probability proportional to $\min_{z \in S} d(x, z)^2$, and add it to S .
3. **Output:** the clustering obtained by the Voronoi partitioning of the data using the centers in S .

[15] showed that Algorithm KMEANS++ is an $\log k$ approximation algorithm for the k -means objective. We say that an algorithm is an α -approximation for a given objective function Φ if for every clustering instance the algorithm outputs a solution of expected cost at most α times the cost of the best solution. The design of approximation algorithms for NP -hard problems has been a fruitful research direction and has led to a wide array of tools and techniques. Formally, [15] show that:

Theorem 1 ([15]). *Let S be the set of centers output by the above algorithm and $\Phi(S)$ be the k -means cost of the clustering obtained using S as the centers. Then $E[\Phi(S)] \leq O(\log k) \text{OPT}$, where OPT is the cost of the optimal k -means solution.*

We would like to point out that in general the output of k -means++ is not a local optimum. Hence it might be desirable in practice to run a few steps of the Lloyd’s method starting from this solution. This could only lead to a better solution.

Subsequent work of [6] introduced a streaming algorithm inspired by the k -means++ algorithm that makes a single pass over the data. They show that if one is allowed to cluster using a little more than k centers, specifically $O(k \log k)$ centers, then one can achieve a constant-factor approximation in expectation to the k -means objective. The approximation guarantee was improved in [5]. Such approximation algorithms which use more than k centers are also known as bi-criteria approximations.

As mentioned earlier, Lloyd’s method can take up to exponential iterations in order to converge to a local optimum. However [13] showed that the method converges quickly on an “average” instance. In order to formalize this, they study the problem under the smoothed analysis framework of [65]. In the smoothed analysis framework the input is generated by applying a small Gaussian perturbation to an adversarial input. [65] showed that the simplex method takes polynomial number of iterations on such smoothed instances. In a similar spirit, [13] showed that for smoothed instances Lloyd’s method runs in time polynomial in n , the number of points and $\frac{1}{\sigma}$, the standard deviation of the Gaussian perturbation. However, these works do not provide any guarantee on the quality of the final solution produced.

We would like to point out that in principle the Lloyd’s method can be extended to the k -median objective. A natural extension would be to replace the mean computation in the Reseeding step with computing the median of a set of points X in the Euclidean space, i.e., a point $c \in \mathbb{R}^d$ such that $\sum_{x \in X} d(x, c)$ is minimized. However this problem turns out to be NP -complete [53]. For this reason, the Lloyd’s method is typically used only for the k -means objective.

3 Properties of the k-means objective

In this section we provide some useful facts about the k-means clustering objective. We will use $\mathcal{C}$ to denote the set of n points which represent a clustering instance. The first fact can be used to show that given a Voronoi partitioning of the data, replacing a given center with the mean of the corresponding partition can never increase the k -means cost.

Fact 2. *Consider a finite set $X \subset \mathbb{R}^d$ and $c = \text{mean}(X)$. For any $y \in \mathbb{R}^d$, we have that, $\sum_{x \in X} d(x, y)^2 = \sum_{x \in X} d(x, c)^2 + |X|d(c, y)^2$.*

Proof. Representing each point in the coordinate notation as $x = (x_1, x_2, \dots, x_d)$, we have that

$$\begin{aligned} \sum_{x \in X} d(x, y)^2 &= \sum_{x \in X} \sum_{i=1}^d |x_i - y_i|^2 \\ &= \sum_{x \in X} \sum_{i=1}^d (|x_i - c_i|^2 + |c_i - y_i|^2 + 2(x_i - c_i)(c_i - y_i)) \\ &= \sum_{x \in X} d(x, c)^2 + |X|d(c, y)^2 + \sum_{i=1}^d 2(c_i - y_i) \sum_{x \in X} (x_i - c_i) \\ &= \sum_{x \in X} d(x, c)^2 + |X|d(c, y)^2 \end{aligned}$$

Here the last equality follows from the fact that for any i , $c_i = \sum_{x \in X} x_i / n$. □

An easy corollary of the above fact is the following,

Corollary 3. *Consider a finite set $X \subset \mathbb{R}^d$ and let $c = \text{mean}(X)$. We have $\sum_{x, y \in X} d(x, y)^2 = 2|X| \sum_{x \in X} d(x, c)^2$.*

Below we prove another fact which will be useful later.

Fact 4. *Let $X \subset \mathbb{R}^d$ be finite set of points. Let $\Delta_1^2(X)$ denote the 1-means cost of X . Given a partition of X into X_1 and X_2 such that $c = \text{mean}(X)$, $c_1 = \text{mean}(X_1)$ and $c_2 = \text{mean}(X_2)$, we have that a) $\Delta_1^2(X) = \Delta_1^2(X_1) + \Delta_1^2(X_2) + \frac{|X_1||X_2|}{|X|}d(c_1, c_2)^2$. and b) $d(c, c_1)^2 \leq \frac{\Delta_1^2(X)|X_2|}{|X||X_1|}$.*

Proof. We can write $\Delta_1^2(X) = \sum_{x \in X_1} d(x, c)^2 + \sum_{x \in X_2} d(x, c)^2$. Using Fact 2 we can write

$$\sum_{x \in X_1} d(x, c)^2 = \Delta_1^2(X_1) + |X_1|d(c, c_1)^2.$$

Similarly, $\sum_{x \in X_2} d(x, c)^2 = \Delta_1^2(X_2) + |X_2|d(c, c_2)^2$. Hence we have

$$\Delta_1^2(X) = \Delta_1^2(X_1) + \Delta_1^2(X_2) + |X_1|d(c, c_1)^2 + |X_2|d(c, c_2)^2.$$

Part (a) follows by substituting $c = \frac{|X_1|c_1 + |X_2|c_2}{|X_1| + |X_2|}$ in the above equation.

From Part (a) we have that

$$\Delta_1^2(X) \geq \frac{|X_1||X_2|}{|X|} d(c_1, c_2)^2.$$

Part (b) follows by substituting $c_2 = \frac{(|X_1| + |X_2|)}{|X_2|} c - \frac{|X_1|}{|X_2|} c_1$ above. $\square$

4 Local search based algorithms

In the previous section we saw that a carefully chosen seeding can lead to a good approximation for the k -means objective. In this section we will see how to design much better (constant factor) approximation algorithms for k -means (as well as k -median). We will describe a very generic approach based on local search. These algorithms work by making local changes to a candidate solution and improving it at each step. They have been successfully used for a variety of optimization problems [7, 28, 36, 40, 58, 61]. Kanungo et. al [47] analyzed a simple local search based algorithm for k -means as described below.

Algorithm LOCAL SEARCH

1. **Initialization:** Choose k data points $\{c_1, c_2, \dots, c_k\}$ arbitrarily from the data set $\mathcal{D}$. Let this set be T . Let $\Phi(T)$ denote the cost of the k -means solution using T as centers, i.e., $\Phi(T) = \sum_{i=1}^n \min_j d^2(x_i, c_j)$. Set $T_{old} = \phi$, $T_{curr} = T$.
2. **While** $\Phi(T_{curr}) < \Phi(T_{old})$,
 - For $x \in T_{curr}$ and $y \in \mathcal{D} \setminus T_{curr}$:
if $\Phi((T_{curr} \setminus \{x\}) \cup \{y\}) < \Phi(T_{curr})$, update $T_{old} = T_{curr}$ and $T_{curr} \leftarrow (T_{curr} \setminus \{x\}) \cup \{y\}$.
3. **Output:** $S = T_{curr}$ as the set of final centers.

We would like to point out that in order to make the above algorithm run in polynomial time, one needs to change the criteria in the while loop to be $\Phi(T_{curr}) < (1 - \epsilon)\Phi(T_{old})$. The running time will then depend polynomially in n and $1/\epsilon$. For simplicity of analysis, we will prove the following theorem for the idealized version of the algorithm with no ϵ .

Theorem 5 ([47]). *Let S be the final set of centers returned by the above procedure. Then, $\Phi(S) \leq 50\text{OPT}$.*

In order to prove the above theorem we start by building up some notation. Let T be the set of k data points returned by the local search algorithm as candidate centers. Let O be the set of k data points which achieve the minimum value of the k -means cost function among all sets of k data points. Note that the centers in O do not necessarily represent the optimal solution as the optimal centers might not be data points. However using the next lemma one can show that using data points as centers is only twice as bad as the optimal solution.

Lemma 6. *Given $\mathcal{C} \subseteq \mathbb{R}^d$, and the optimal k -means clustering of $\mathcal{C}$, $\{C_1, C_2, \dots, C_k\}$, there exists a set S of k data points such that $\Phi(S) \leq 2\text{OPT}$.*

Proof. For a given set $\mathcal{C} \subseteq \mathbb{R}^d$, let Δ_1^2 represent the 1-means cost of $\mathcal{C}$. From Fact 2 it is easy to see that this cost is achieved by choosing the mean of $\mathcal{C}$ as the center. In order to prove the above lemma it is enough to show that for each optimal cluster C_i with mean c_i , there exists a data point $x_i \in C_i$ such that $\sum_{x \in C_i} d(x, x_i)^2 \leq 2\Delta_1^2(C_i)$. Let x_i be the data point in C_i which is closest to c_i . Again using Fact 2 we have $\sum_{x \in C_i} d(x, x_i)^2 = \Delta_1^2(C_i) + |C_i|d(x, c_i)^2 \leq 2\Delta_1^2(C_i)$. $\square$

Hence it is enough to compare the cost of the centers returned by the algorithm to the cost of the optimal centers using data points. In particular, we will show that $\Phi(T) \leq 25\Phi(O)$. We start with the simple observation that by the property of the local search algorithm, for any $t \in T$, and $o \in O$, swapping t for o results in an increase in cost. In other words

$$\Phi(T - t + o) - \Phi(T) \geq 0 \quad (4.1)$$

The main idea is to add up Equation 4.1 over a carefully chosen set of swaps $\{o, t\}$ to get the desired result. In order to describe the set of swaps chosen we start by defining a cover graph

Definition 2. *A cover graph is a bipartite graph with the centers in T on one side and the centers in O on the other side. For each $o \in O$, let t_o be the point in T which is closest to o . The cover graph contains edges of the form o, t_o for all $o \in O$.*

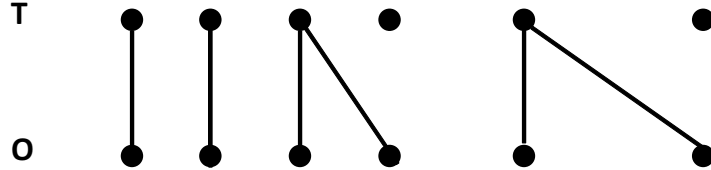


Figure 2: An example cover graph.

Next we use the cover graph to generate the set of useful swaps. For each $t \in T$ which has degree 1 in the cover graph, we output the swap pair $\{t, o\}$ where o is the point connected to t . Let T' be the degree 0 vertices in the cover graph. We pair the remaining vertices $o \in O$ with the vertices in T' such that each vertex in O has degree 1 and each vertex in T' has degree at most 2. To see that such a pairing will exist notice that for any $t \in T$ of degree $k > 1$ there will exist $k - 1$ distinct zero vertices in T ; these vertices can be paired to vertices in O connected to t maintaining the above property. We then output all the edges in this pairing as the set of useful swaps.

4.1 Bounding the cost of a swap

Consider a swap $\{o, t\}$ output by using the cover graph. We will apply Equation 4.1 to this pair. We will explicitly define a clustering using centers in $T - t + o$ and upper bound its cost.

We will then use the lower bound of $\Phi(T)$ from Equation 4.1 to get the kind of equations we want to sum up over. Let the clustering given by centers in T be $C_1, C_2, \dots, C_k$. Let C_o^* be the cluster corresponding to center o in the optimal clustering given by O . Let o_x be the closest point in O to x . Similarly let t_x be the closest point in T to x . The key property satisfied by any output pair $\{o, t\}$ is the following

Fact 7. *Let $\{o, t\}$ be a swap pair output using the cover graph. Then we have that for any $x \in C_t$ either $o_x = o$ or $t_{o_x} \neq t$.*

Proof. Assume that for some $x \in C_t$, $o_x = o' \neq o$. By the procedure used to output swap pairs we have that t has degree 1 or 0 in the cover graph. In addition, if t has degree 1 then $t_o = t$. In both the cases we have that $t_{o'} \neq t$. $\square$

Next we create a new clustering by swapping o for t and assigning all the points in C_o^* to o . Next we reassign points in $C_t \setminus C_o^*$. Consider a point $x \in C_t \setminus C_o^*$. Clearly $o_x \neq o$. Let t_{o_x} be the point in T which is connected to o_x in the cover graph. We assign x to t_{o_x} . One needs to ensure here that $t_{o_x} \neq t$ which follows from Fact 7. From Equation 4.1 the increase in cost due to this reassignment must be non-negative. In other words we have

$$\sum_{x \in C_o^*} (d(x, o)^2 - d(x, t_x)^2) + \sum_{x \in C_t \setminus C_o^*} (d(x, t_{o_x})^2 - d(x, t)^2) \geq 0 \quad (4.2)$$

We will add up Equation 4.2 over the set of all good swaps.

4.2 Adding it all up

In order to sum up over all swaps notice that in the first term in Equation 4.2 every point $x \in \mathcal{C}$ appears exactly once by being in C_o^* for some $o \in O$. Hence the sum over all swaps of the first term can be written as $\sum_{x \in \mathcal{C}} (d(x, o_x)^2 - d(x, t_x)^2)$. Consider the second term in Equation 4.2. We have that $(d(x, t_{o_x})^2 - d(x, t)^2) \geq 0$ since x is in C_t . Hence we can replace the second summation over all $x \in C_t$ without affecting the inequality. Also every point $x \in \mathcal{C}$ appears at most twice in the second term by being in C_t for some $t \in T$. Hence the sum over all swaps of the second term is at most $\sum_{x \in \mathcal{C}} (d(x, t_{o_x})^2 - d(x, t_x)^2)$. Adding these up and rearranging we get that

$$\Phi(O) - 3\Phi(T) + 2R \geq 0 \quad (4.3)$$

Here $R = \sum_{x \in \mathcal{C}} d(x, t_{o_x})^2$.

In the last part we will upper bound the quantity R . R represents the cost of assigning every point x to a center in T but not necessarily the closest one. Hence, $R \geq \Phi(T) \geq \Phi(O)$. However we next show that this reassignment cost is not too large.

Notice that R can also be written as $\sum_{o \in O} \sum_{x \in C_o^*} d(x, t_o)^2$. Also $\sum_{x \in C_o^*} d(x, t_o)^2 = \sum_{x \in C_o^*} d(x, o)^2 + |C_o^*|d(o, t_o)^2$. Hence we have that $R = \sum_{o \in O} \sum_{x \in C_o^*} (d(x, o)^2 + d(o, t_o)^2)$.

Also note that $d(o, t_o) \leq d(o, t_x)$ for any x . Hence

$$\begin{aligned} R &\leq \sum_{o \in O} \sum_{x \in C_o^*} (d(x, o)^2 + d(o, t_x)^2) \\ &= \sum_{x \in \mathcal{C}} (d(x, o_x)^2 + d(o_x, t_x)^2) \end{aligned}$$

Using triangle inequality we know that $d(o_x, t_x) \leq d(o_x, x) + d(x, t_x)$. Substituting above and expanding we get that

$$R \leq 2\Phi(O) + \Phi(T) + 2 \sum_{x \in \mathcal{C}} d(x, o_x) d(x, t_x) \quad (4.4)$$

The last term in the above equation can be bounded using Cauchy-Schwarz inequality as $\sum_{x \in \mathcal{C}} d(x, o_x) d(x, t_x) \leq \sqrt{\Phi(O)} \sqrt{\Phi(S)}$. So we have that $R \leq 2\Phi(O) + \Phi(T) + 2\sqrt{\Phi(O)} \sqrt{\Phi(S)}$. Substituting this in Equation 4.3 and solving we get the desired result that $\Phi(T) \leq 25\Phi(O)$. Combining this with Lemma 6 proves Theorem 5.

A natural generalization of Algorithm LOCAL SEARCH is to swap more than one centers at each step. This could potentially lead to a much better local optimum. This multi-swap scheme was analyzed by [47] and using a similar analysis as above one can show the following

Theorem 8. *Let S be the final set of centers by the local search algorithm which swaps upto p centers at a time. Then we have that $\Phi(S) \leq 2(3 + \frac{2}{p})^2 \text{OPT}$, where OPT is the cost of the optimal k -means solution.*

For the case of k -median the same algorithm and analysis gives [16]

Theorem 9. *Let S be the final set of centers by the local search algorithm which swaps upto p centers at a time. Then we have that $\Phi(S) \leq (3 + \frac{2}{p}) \text{OPT}$, where OPT is the cost of the optimal k -median solution.*

This approximation factor for k -median has recently been improved to $(1 + \sqrt{3} + \epsilon)$ [50]. For the case of k -means in Euclidean space [48] give an algorithm which achieves a $(1 + \epsilon)$ approximation to the k -means objective for any constant $\epsilon > 0$. However the runtime of the algorithm depends exponentially in k and hence it is only suitable for small instances.

5 Clustering of *stable* instances

In this part of the chapter we delve into some of the more modern research in the theory of clustering. In recent past there has been an increasing interest in designing clustering algorithms that enjoy strong theoretical guarantees on non-worst case instance. This is of significant interest for two reasons: a) From a theoretical point of view, this helps us understand and characterize the class of problems for which one can get optimal or close